 

# PEMBROKE, ON, Canada

WMO: **711960**

Lat: **45.8603N** Lon: **77.2525W** Elev: **162** StdP: **99.39** Time Zone: **-5.00 (NAE)** Period: **94-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **1** | **-24.8** | **-21.1** | **-28.4** | **0.3** | **-24.4** | **-24.9** | **0.4** | **-20.4** | **24.8** | **-1.1** | **23.1** | **-5.6** | **2.1** | **300** | **0.738** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **7** | **9.3** | **28.3** | **19.3** | **26.3** | **18.9** | **24.4** | **18.0** | **21.4** | **25.6** | **20.3** | **24.3** | **19.2** | **22.9** | **3.4** | **230** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **20.0** | **15.0** | **23.8** | **18.8** | **13.9** | **22.4** | **17.8** | **13.0** | **21.2** | **63.0** | **25.8** | **59.2** | **24.3** | **55.4** | **22.9** | **26.1** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **21.1** | **18.3** | **15.7** | DB | **-22.1** | **28.9** | **8.9** | **2.9** | **-28.5** | **31.0** | **-33.7** | **32.7** | **-38.7** | **34.3** | **-45.2** | **36.4** | (4) |
| (5) | | | | WB | **-24.2** | **22.7** | **8.7** | **1.8** | **-30.4** | **23.9** | **-35.5** | **25.0** | **-40.4** | **26.0** | **-46.7** | **27.3** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | **Temperatures, Degree-Days and Degree-Hours** | | DBAvg | **5.1** | **-7.4** | **-6.9** | **-3.4** | **2.3** | **8.5** | **13.0** | **17.6** | **17.1** | **13.5** | **7.9** | **2.1** | **-3.4** | (6) |
| (7) | | | DBStd | **9.99** | **7.03** | **6.12** | **4.84** | **3.73** | **5.31** | **5.15** | **3.90** | **3.10** | **3.76** | **3.76** | **4.46** | **6.04** | (7) |
| (8) | | | HDD10.0 | **2535** | **538** | **472** | **414** | **233** | **96** | **28** | **2** | **0** | **8** | **89** | **239** | **416** | (8) |
| (9) | | | HDD18.3 | **4894** | **797** | **705** | **673** | **481** | **309** | **172** | **60** | **60** | **152** | **324** | **488** | **674** | (9) |
| (10) | | | CDD10.0 | **763** | **0** | **0** | **0** | **3** | **49** | **118** | **237** | **219** | **113** | **23** | **1** | **0** | (10) |
| (11) | | | CDD18.3 | **81** | **0** | **0** | **0** | **0** | **4** | **12** | **37** | **21** | **7** | **0** | **0** | **0** | (11) |
| (12) | | | CDH23.3 | **632** | **0** | **0** | **0** | **1** | **46** | **111** | **309** | **125** | **39** | **2** | **0** | **0** | (12) |
| (13) | | | CDH26.7 | **143** | **0** | **0** | **0** | **0** | **9** | **25** | **79** | **23** | **7** | **0** | **0** | **0** | (13) |
| (14) | **Wind** | | WSAvg | **6.2** | **8.1** | **7.6** | **6.8** | **6.1** | **5.2** | **5.4** | **4.6** | **5.1** | **5.7** | **6.1** | **7.3** | **6.8** | (14) |
| (15) | **Precipitation** | | PrecAvg | **777** | **51** | **41** | **44** | **58** | **68** | **73** | **85** | **81** | **76** | **73** | **64** | **60** | (15) |
| (16) | | | PrecMax | **950** | **92** | **80** | **129** | **104** | **103** | **136** | **164** | **130** | **137** | **117** | **104** | **92** | (16) |
| (17) | | | PrecMin | **641** | **18** | **16** | **13** | **17** | **33** | **17** | **20** | **21** | **36** | **12** | **20** | **18** | (17) |
| (18) | | | PrecStd | **75** | **21** | **16** | **26** | **24** | **21** | **27** | **40** | **21** | **29** | **29** | **21** | **19** | (18) |
| (19) | **Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures** | 0.4% | DB | | **9.4** | **7.2** | **9.0** | **18.6** | **27.9** | **29.6** | **31.7** | **29.2** | **27.2** | **21.3** | **14.5** | **9.8** | (19) |
| (20) | | | MCWB | | **8.9** | **6.1** | **7.0** | **10.3** | **17.9** | **19.6** | **21.3** | **18.9** | **20.7** | **16.1** | **12.0** | **8.9** | (20) |
| (21) | | 2% | DB | | **5.2** | **4.2** | **6.2** | **13.6** | **24.2** | **26.4** | **28.9** | **26.4** | **24.0** | **17.9** | **12.3** | **7.0** | (21) |
| (22) | | | MCWB | | **4.5** | **3.4** | **3.7** | **8.4** | **15.4** | **18.2** | **19.8** | **19.2** | **18.8** | **14.8** | **10.5** | **6.3** | (22) |
| (23) | | 5% | DB | | **2.2** | **2.4** | **4.2** | **10.3** | **21.0** | **23.8** | **26.8** | **24.4** | **21.6** | **15.8** | **10.1** | **4.5** | (23) |
| (24) | | | MCWB | | **1.4** | **1.5** | **2.3** | **6.5** | **14.2** | **17.0** | **19.3** | **18.5** | **17.9** | **13.2** | **8.8** | **3.7** | (24) |
| (25) | | 10% | DB | | **0.6** | **1.0** | **2.7** | **8.0** | **17.8** | **21.4** | **24.7** | **22.5** | **19.4** | **13.7** | **8.1** | **2.9** | (25) |
| (26) | | | MCWB | | **-0.3** | **0.1** | **1.2** | **4.8** | **12.6** | **16.1** | **18.7** | **17.8** | **16.5** | **11.7** | **6.9** | **2.1** | (26) |
| (27) | **Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures** | 0.4% | WB | | **9.1** | **6.0** | **7.7** | **13.9** | **20.0** | **21.7** | **23.4** | **21.7** | **22.0** | **18.0** | **13.0** | **8.9** | (27) |
| (28) | | | MCDB | | **9.4** | **6.5** | **8.7** | **17.0** | **25.0** | **26.4** | **29.3** | **25.9** | **25.9** | **19.7** | **13.7** | **9.4** | (28) |
| (29) | | 2% | WB | | **4.5** | **3.6** | **4.3** | **9.6** | **17.6** | **20.0** | **21.7** | **20.5** | **20.0** | **15.4** | **10.9** | **6.2** | (29) |
| (30) | | | MCDB | | **5.1** | **4.2** | **5.3** | **12.1** | **21.9** | **23.9** | **26.1** | **24.1** | **23.1** | **17.2** | **11.9** | **6.9** | (30) |
| (31) | | 5% | WB | | **1.6** | **1.7** | **2.6** | **7.4** | **15.3** | **18.2** | **20.5** | **19.5** | **18.2** | **13.5** | **8.9** | **3.8** | (31) |
| (32) | | | MCDB | | **2.1** | **2.2** | **3.8** | **10.1** | **19.3** | **22.1** | **25.0** | **22.9** | **20.7** | **15.3** | **10.1** | **4.3** | (32) |
| (33) | | 10% | WB | | **-0.2** | **0.2** | **1.3** | **5.0** | **13.1** | **16.6** | **19.3** | **18.5** | **16.7** | **11.8** | **7.1** | **2.2** | (33) |
| (34) | | | MCDB | | **0.5** | **0.8** | **2.5** | **7.7** | **18.0** | **20.7** | **23.6** | **21.4** | **19.1** | **13.5** | **8.1** | **2.7** | (34) |
| (35) | **Mean Daily Temperature Range** | | MDBR | | **7.5** | **7.3** | **6.8** | **6.6** | **8.4** | **8.5** | **9.3** | **8.0** | **7.8** | **6.4** | **6.0** | **5.6** | (35) |
| (36) | | 5% DB | MCDBR | | **9.3** | **7.9** | **8.9** | **12.4** | **14.9** | **13.0** | **13.6** | **12.0** | **12.0** | **10.9** | **9.2** | **7.2** | (36) |
| (37) | | | MCWBR | | **9.5** | **7.4** | **7.3** | **8.2** | **7.2** | **6.2** | **6.3** | **6.1** | **7.0** | **7.7** | **8.1** | **7.6** | (37) |
| (38) | | 5% WB | MCDBR | | **9.3** | **7.5** | **7.7** | **10.8** | **11.9** | **10.8** | **11.3** | **9.8** | **10.2** | **9.6** | **8.4** | **6.8** | (38) |
| (39) | | | MCWBR | | **9.5** | **7.3** | **6.9** | **8.3** | **7.0** | **6.3** | **6.1** | **5.9** | **6.7** | **7.6** | **8.2** | **7.3** | (39) |
| (40) | **Clear-Sky Solar Irradiance** | taub | | | **0.268** | **0.285** | **0.318** | **0.352** | **0.370** | **0.410** | **0.406** | **0.393** | **0.369** | **0.342** | **0.308** | **0.273** | (40) |
| (41) | | taud | | | **2.359** | **2.301** | **2.351** | **2.383** | **2.388** | **2.294** | **2.322** | **2.355** | **2.417** | **2.477** | **2.498** | **2.397** | (41) |
| (42) | | Ebn at Noon | | | **844** | **895** | **912** | **906** | **897** | **858** | **857** | **855** | **847** | **822** | **793** | **797** | (42) |
| (43) | | Edh at Noon | | | **73** | **95** | **105** | **112** | **116** | **128** | **123** | **115** | **99** | **81** | **66** | **64** | (43) |
| (44) | **All-Sky Solar Radiation** | RadAvg | | | **1.35** | **2.31** | **3.67** | **4.55** | **5.17** | **5.69** | **5.65** | **4.94** | **3.85** | **2.26** | **1.34** | **0.98** | (44) |
| (45) | | RadStd | | | **0.19** | **0.27** | **0.32** | **0.39** | **0.46** | **0.43** | **0.47** | **0.28** | **0.36** | **0.23** | **0.15** | **0.12** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (19 neighbors) | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (47) |

Nomenclature: See separate page